

TiO₂ Scattering Phantom for Optical Coherence Tomography

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1. Purpose and Measurement Metrics

The purpose of these TiO₂-based optical scattering phantoms is to provide stable, well-characterized physical reference standards for optical coherence tomography (OCT). OCT is not limited to structural imaging. In addition to visualizing tissue morphology, quantitative analysis of the OCT signal enables estimation of optical properties, including the optical attenuation coefficient, which is related to tissue microstructure, organization, and composition and may provide additional diagnostic information^{1,2}. Quantitative attenuation analysis has therefore been investigated as a potential approach for tissue differentiation and for improving the diagnostic capability of OCT.

However, accurate attenuation measurement remains challenging. Conventional OCT-based attenuation estimates can be biased by simplifying model assumptions, the confocal point-spread function, sensitivity roll-off, and the contribution of multiply scattered photons, all of which can reduce the accuracy of optical-property estimation³. These effects are particularly significant in highly scattering media, where multiple scattering and depth-dependent signal distortions become more pronounced. For this reason, reliable physical reference standards are needed. The TiO₂-based phantoms described here are designed to support validation of optical attenuation coefficient (OAC) quantification methods by providing controlled scattering references with independently measured optical properties. OCT-derived attenuation values can therefore be evaluated against reference measurements obtained using established optical characterization techniques.

The optical properties of biological tissues are commonly described by the reduced scattering coefficient, μ'_s ; absorption coefficient, μ_a ; scattering coefficient, μ_s ; and anisotropy factor, g ⁴. These parameters depend on tissue type, composition, and wavelength. In OCT, the optical attenuation coefficient is determined by both absorption and scattering and is commonly modeled as the sum of these contributions. In this protocol, integrating-sphere measurements combined with inverse adding-doubling analysis are used to quantify μ'_s and μ_a ^{5,6}. A simple and robust low-coherence interferometry method is then applied to quantify μ_s . The anisotropy factor, g , is subsequently calculated from the relationship:

$$\mu'_s = (1 - g)\mu_s$$

The intended applications of these phantoms are specific and well defined. They are intended for OCT system performance assessment, repeatability testing, and comparison of attenuation-measurement methods. They are not intended to serve as complete biological tissue models, nor are they clinical diagnostic devices. Rather, they are controlled optical scattering references designed to support the validation and standardization of OCT-based quantitative measurement methods.

2. Physical Reference Object

The physical reference object in this phantom package is cartridge-format TiO₂-based optical scattering phantoms. The phantom is intended to serve as the physical reference object for OCT imaging. Each completed phantom is assigned a unique serial number with a model number. The serial number and the model number are printed on the side of the cartridge and is used to identify the individual phantom throughout fabrication, quality control, storage, use, and future reference.

Model NO.: OCTP820 -TIO2

SN: S###-G##- B##-YYMMDD

OCTP820 = COT phantom calibrated at 820nm

T = TiO₂ scattering phantom

S### = scattering coefficient μ_s in mm⁻¹ ×10

G## = anisotropy×100

B## = batch number

YYDD = batch date

Examples:

$\mu s = 2.0 \text{ mm}^{-1}$ $g = 0.90$ batch = 1 date = 2026-05-15 scatters = TiO_2	OCT820-TIO2 SN: S020-G90-B01-260515
$\mu s = 0.5 \text{ mm}^{-1}$ $g = 0.85$ batch = 2 date = 2026-06-15 scatters = TiO_2	OCT820-TIO2 SN: S005-G85-B02-260615

The phantoms are stored and transported in a protective container to reduce mechanical damage, contamination, and damage to the optical window. No separate mounting fixture is required for routine storage or handling. If a dedicated holder or alignment fixture is used during OCT imaging, that fixture is stored with the phantom and labeled with the corresponding phantom serial number.

3. Fabrication Protocol

This part describes fabrication of a TiO_2 -doped PDMS optical scattering phantom assembled in a 3D-printed cartridge with a cover glass, sealed sample chamber, buffer chamber, inlet port, epoxy sealing channel, and protective cartridge enclosure. The procedure covers cartridge preparation, scattering-PDMS preparation, filling, vacuum degassing, curing, final enclosure assembly, and pre-release fabrication inspection.

4. Fabrication Workflow Overview

The fabrication workflow shall be followed in the order shown in Figure 1.

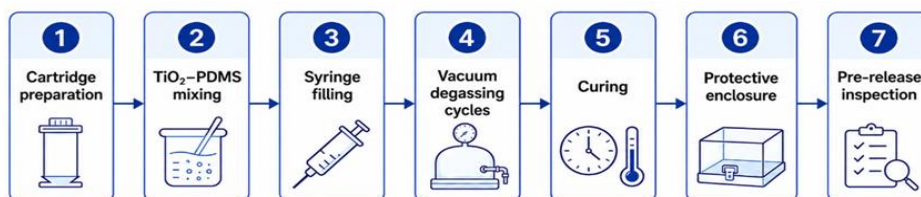


Figure 1. The fabrication flow of the phantom

4.1. Cartridge Components

Figure 2 identifies the key cartridge components and supporting fabrication parts.

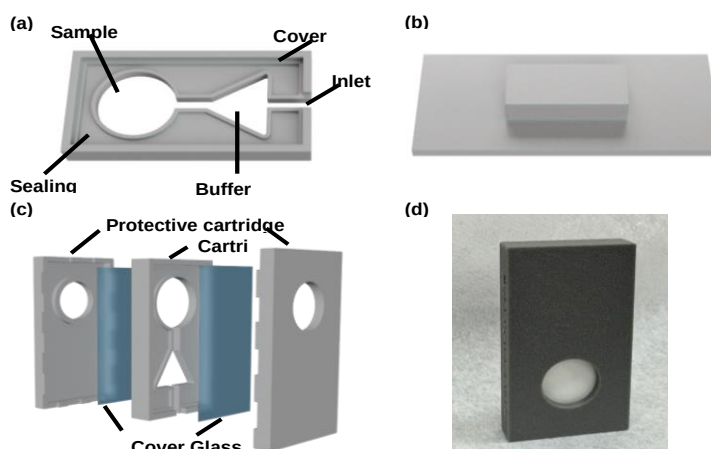


Figure 2. Cartridge-based TiO_2 optical scattering phantom components. A) 3D-printed cartridge schematic with sample chamber, cover glass seat, inlet port, sealing channel, and buffer chamber. B) 3D-printed glass weight used during cover-glass sealing. C) Exploded view of cartridge and protective enclosure. D) Placeholder image of final phantom cartridge.

4.2. Materials and equipment

Prepare the following materials and equipment before fabrication begins.

- 3D printer (P1S, Bambu)
- 3D-printed cartridge base
- Cover glass (24 mm x 50 mm) (CS-24X50-100, Amscope)
- Sylgard 184 PDMS elastomer kit (base + curing agent) (68083-19-2/68988-89-6, DOW)
- Gorilla Epoxy (4200102, The Gorilla Glue Company)
- 99.9% 100 nm titanium dioxide (TiO₂) nano powder (US3535, US Research Nanomaterials)
- Syringe with flat-tip needle
- Disposable weighing boats
- Mixing silicone containers
- Magnetic stir plate (PC-420D, Corning)
- Magnetic stir bar
- Ultrasonic bath (230HT, Zokop)
- Vacuum chamber and vacuum pump (RS-1, 1 Stage)
- 3D-printed glass weight
- Protective cartridge enclosure
- Acetone-moistened wipe for excess epoxy removal during final assembly

4.3. Cartridge preparation

1. Mix equal volumes of epoxy resin and hardener in a clean disposable container, such as a weighing boat.
2. Pour 0.825 mL epoxy to the sealing channel of the 3D-printed cartridge evenly.
3. Place the cover glass into the cover glass seat of the cartridge.
4. Place a 3D-printed weight on top of the cover glass to apply uniform pressure and ensure that the glass sits flat, then remove the weight.
5. Repeat the epoxy filling step to seal the sealing channels on both sides of the cartridge.
6. Allow the epoxy to cure at room temperature for 24 hours.

4.4. Scattering PDMS sample preparation

7. Weigh the desired amount of PDMS base into a clean mixing silicone container and record the mass.
8. Weigh the required amount of TiO₂ powder in a weighing boat according to the target scattering concentration and record the mass.
9. Add the TiO₂ powder to the PDMS base.
10. Mix the suspension using a magnetic stir plate and stir bar for 20 minutes at 200 rpm to disperse the particles.
11. Place the mixture in an ultrasonic water bath (40KHz/120W) for 20 minutes to improve TiO₂ dispersion.
12. Add PDMS curing agent to the mixture using a 10:1 base-to-curing-agent weight ratio.
13. Mix again on the magnetic stir plate for 20 minutes at 200 rpm.

4.5. Phantom filling and degassing

14. Draw the PDMS-TiO₂ mixture into a syringe.
15. Inject the mixture into the cartridge through the PDMS inlet until the buffer chamber is filled to approximately two-thirds capacity.
16. Place the cartridge in a vacuum chamber.
17. Slowly reduce the chamber pressure using the vacuum pump.
18. When the PDMS begins to foam and bubbles expand, monitor the chamber carefully.
19. Temporarily close the vacuum valve if foam approaches the cartridge opening.
20. Wait until the bubbles collapse.
21. Slowly vent the chamber to atmospheric pressure.

22. Allow the cartridge to rest for 5 minutes.
23. Repeat the vacuum cycle until no bubbles form at approximately 29 inHg vacuum.

4.6. Phantom curing

24. Place the cartridge upright at room temperature for 24 hours to fully cure the PDMS.

4.7. Final assembly

25. Install the protective enclosure over the cartridge.
26. Apply epoxy adhesive along the enclosure seams to seal the cartridge.
27. Remove any excess epoxy that squeezes out from the seams using a wipe moistened with acetone.
28. Once the excess epoxy is cleaned and the enclosure is secured, the device can be used immediately without waiting for full epoxy curing.

5. Quantification

5.1. Instrumentation

- Integral spheres: General Purpose Integrating Sphere from Labsphere
- Spectrometer: LR1 from ASEQ Instruments
- Laser: SLD371 from Superlum Ireland
- Low coherent interferometer: In-house built 850nm spectral domain interferometer

5.2. Software

- Inverse Adding-Doubling (IAD) (<https://github.com/scottprahl/iad>)
- OCTSharp (<https://github.com/OCTSharpImaging/OCTSharp>)

5.3. Method: Reduced scattering and absorption coefficient measurement

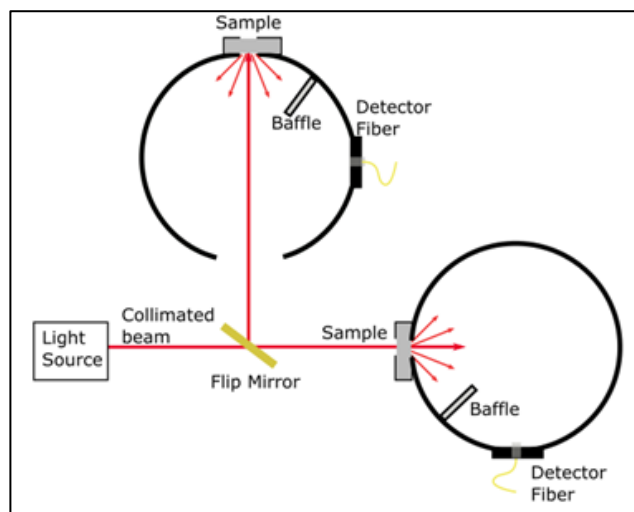


Figure 3. The Integral sphere system used to quantify the reduced scattering coefficient μ'_s and absorption coefficient μ_a of the phantoms.

The reduced scattering coefficient (μ'_s) and attenuation coefficient (μ_a) of each TiO_2 -PDMS phantom are determined using integrating spheres followed by Inverse Adding-Doubling (IAD) analysis using the optical setup shown in Figure 3. The flip mirror is used to direct the beam to either the reflectance integrating sphere or the transmittance integrating sphere. For the reflectance measurement, the sample is mounted at the sample port of the integrating sphere, and the diffusely reflected light is collected by the detector fiber. Before sample measurement, dark measurements and reference measurements are collected for background subtraction and instrument normalization. The corrected diffuse reflectance and total transmittance values are then entered into the IAD calculation together with the sample thickness of 3 mm, measurement wavelength of 824.4 nm, refractive index of the sample of 1.41 (PDMS), refractive index of

the surrounding medium of 1.51 (Glass slide), and integrating-sphere geometry. The IAD algorithm estimates the sample optical properties as μ'_s (mm^{-1}) and μ_a (mm^{-1}) by iteratively solving the radiative transport problem.

5.4. Method: Attenuation measurement

The attenuation coefficient (μ_t) shall be measured as an independent parameter to support the IAD calculation. Conventional collimated transmission, shown in Fig. 4(a), estimates ballistic transmittance by measuring light transmitted through the sample. This method is sensitive to sample thickness, detector dynamic range, and

collection geometry because forward-scattered light may reach the detector and cause underestimation of μ_t . Therefore, it shall not be used as the primary attenuation measurement for these scattering phantoms⁵.

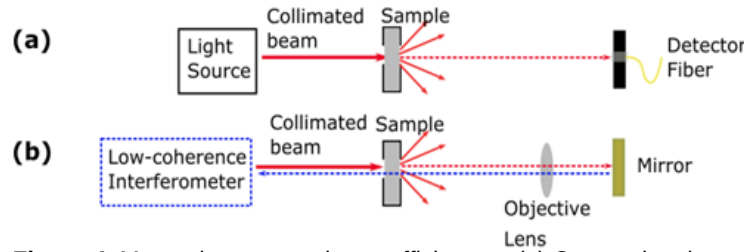


Figure 4. Measuring attenuation coefficient, μ_t . (a) Conventional transmission method using a collimated beam and detector fiber. (b) Low-coherence interferometry (LCI)-based method using back reflected collimated beam from a mirror

In this protocol, μ_t has been measured using the low-coherence interferometer (LCI) configuration shown in Fig. 4(b). The LCI beam passes through the sample, reflects from a mirror, and passes through the sample a second time before returning to the interferometer. The mirror interference peak is used as the measurement signal. Because LCI has coherence gating, the detected signal is dominated by ballistic or least-scattered photons, reducing the contribution of multiply scattered light. The sample is placed in the collimated beam before the objective lens to eliminate confocal-gate effects during attenuation measurement.

5.5. Measured results of the phantoms

To determine the anisotropy factor, g , of the TiO_2 phantoms, we fabricated three phantoms with different TiO_2 concentrations, as listed in Table 1. Because absorption by the phantoms at approximately 800 nm is very low, two orders less than scattering coefficients, it is reasonable to assume that $\mu_s \approx \mu_t$, where μ_t was measured using the LCI-based method. The resulting μ_s values were then used as inputs to the IAD algorithm, together with the data acquired using the double integrating spheres, to calculate g .

Table 1. Optical properties of TiO_2 phantoms used to calculate the anisotropy factor g .

Concentration (mg/g)	μ'_s (mm ⁻¹)	μ_s (mm ⁻¹)	g
0.5	0.3 ± 0.0027	0.88 ± 0.0032	0.66 ± 0.0033
0.75	0.44 ± 0.006	1.31 ± 0.027	0.66 ± 0.0083
1	1.88 ± 0.019	5.48 ± 0.053	0.66 ± 0.0048

Using the determined anisotropy factor, g , we fabricated five phantoms with different TiO_2 concentrations and quantified their optical properties. All measured data are listed in Table 2, together with the corresponding serial numbers. The linear relationship between TiO_2 concentration and the scattering coefficient, μ_s , is plotted in Fig. 5. The OCT images of these phantoms are shown in Fig.6. The cover glass thickness is 190 μm , and the material is BK7 glass with a refractive index of approximately 1.51 at 800 nm. These cover-glass boundaries can be used as the reference for image-dimension verification and prevent the specular reflection from the phantom surface.

Table 2. Optical properties of TiO_2 phantoms computed from double integrating sphere measurements.

Concentration (mg/g)	Serial Number	μ'_s (mm ⁻¹)	μ_a (mm ⁻¹)	μ_s (mm ⁻¹)	g
1	S052-G066-B01-260312	1.76 ± 0.018	0.014 ± 0.0006	5.18 ± 0.053	0.66
5	S137-G066-B01-260313	4.64 ± 0.039	0.0089 ± 0.00028	13.65 ± 0.11	0.66
10	S217-G066-B01-260314	7.39 ± 0.14	0.0082 ± 0.00051	21.73 ± 0.42	0.66
15	S315-G066-B01-260315	10.73 ± 0.15	0.0075 ± 0.00035	31.55 ± 0.44	0.66
20	S414-G066-B01-260316	14.09 ± 0.14	0.0041 ± 0.00043	41.43 ± 0.41	0.66

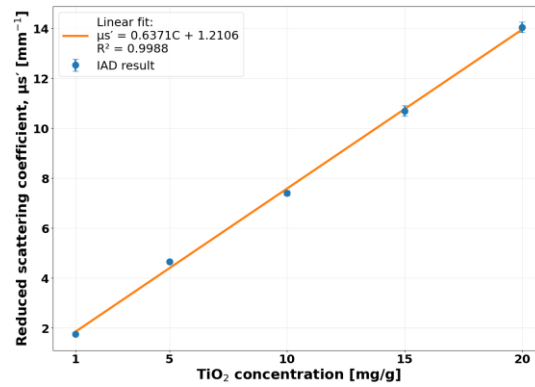


Figure 5. Reduced scattering coefficient as a function of TiO₂ concentration

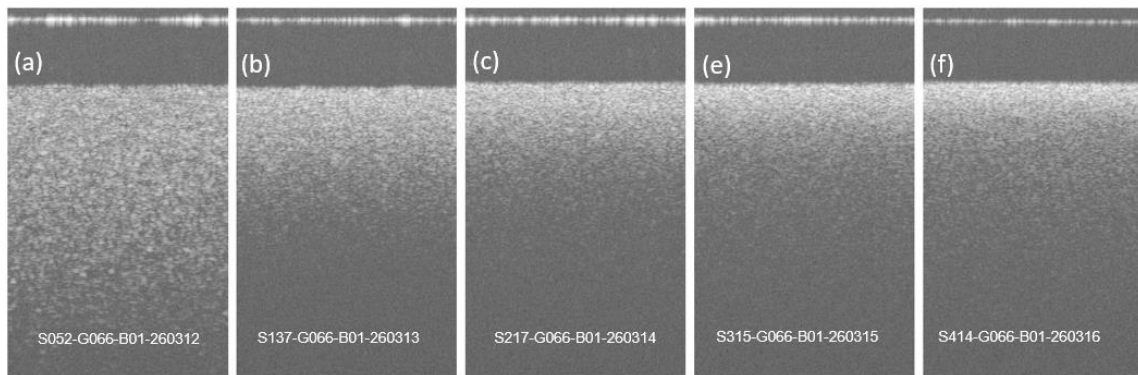


Figure 6. OCT images of the phantoms list in Table 5

6. Operation and Handling

6.1. Operation

Use the phantom only for research and validation of optical imaging/OCT systems unless a separate regulatory or clinical-use document is approved. Handle the cartridge carefully to avoid contamination of the cover glass or optical window. Before use, inspect the phantom for cracks, leakage, delamination, large bubbles, surface contamination, or label damage. Do not use the phantom if leakage, broken glass, optical-window contamination, or unreadable labeling is observed. During imaging, place the phantom under the laser beam using the approved OCT sample holder or stage. The phantom may be placed at a slight angle relative to the illumination optical axis to reduce strong specular reflection from the cover glass.

The phantom should be placed directly beneath the OCT laser beam, with the beam passing through the optical window. To avoid strong specular reflection from the phantom surface, several angled fixtures with different tilt angles were provided for positioning the phantoms.

6.2. Permitted use

- OCT imaging performance evaluation, repeatability testing, and comparative imaging studies using controlled scattering media.
- Bench testing under controlled laboratory conditions.

6.3. Use limitations

- Not intended for direct patient contact.
- Not intended for sterile use.
- Not a clinical diagnostic device.

- Do not expose to extreme heat, freezing conditions, direct sunlight, abrasive cleaning, or unapproved solvents.

7. Environmental Sensitivity and Monitoring

7.1. Fabrication and curing environment

- Recommended fabrication room temperature: 20-25 deg C
- Recommended relative humidity: below 60 percent.
- Fabricate and cure in a clean, dust-minimized area.
- Avoid vibration during curing and protect the sample from dust
- Phantoms shall be inspected before use for cracks, delamination, large bubbles, surface contamination, or other damage.

7.2. Shelf time

Two replicate phantom cartridges are stored under room-temperature conditions and measured weekly to monitor stability over time as shown in r. The weekly monitoring data will be maintained as part of the phantom documentation and updated in the protocol as data become available through GitHub. Until sufficient stability data are collected, no final shelf-life claim is assigned.

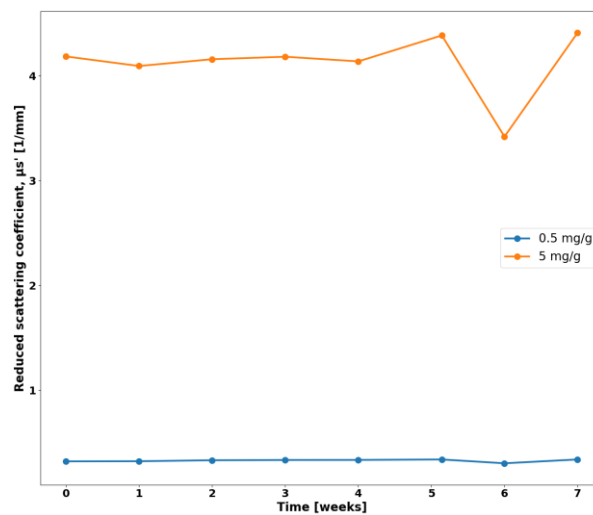


Figure 7. Long-term stability monitoring of the reduced scattering coefficient (μ_s') for TiO_2 phantoms measured by double integrating sphere. (a) 0.5 mg/g TiO_2 phantom (b) 5 mg/g TiO_2 phantom.

8. Safety Components

Safety documentation must be maintained with the phantom package. Current SDS documents should be available to personnel before fabrication begins.

8.1. Required SDS files attached with documentation

- Titanium dioxide powder
- Sylgard 184 PDMS base
- Sylgard 184 PDMS curing agent
- Two-part epoxy resin and hardener

8.2. Personal protective equipment and safe handling

- Wear a lab coat, nitrile gloves, face mask, and safety glasses during all fabrication steps.
- Handle dry TiO_2 powder in a fume hood or other approved local exhaust system to minimize inhalation risk.
- Avoid generating TiO_2 dust clouds; do not blow or dry sweep TiO_2 powder.
- Avoid skin and eye contact with PDMS curing agent, uncured epoxy, acetone, and uncured PDMS mixtures.

8.3. Spill and accident response

- TiO₂ powder spill: dampen powder or use an approved HEPA-filtered vacuum; do not dry sweep.
- Uncured PDMS or epoxy spill: contain and absorb with compatible inert material, then dispose as chemical waste according to institutional policy.
- Skin contact: wash with soap and water and remove contaminated clothing.
- Eye contact: Rinse cautiously with water for several minutes and seek medical attention if irritation persists.
- Inhalation of TiO₂ dust: move to fresh air and seek medical attention if symptoms occur.

8.4. Waste disposal

- Cured PDMS and cured epoxy may be handled as solid waste only if permitted by institutional policy.
- Uncured PDMS, curing agent, uncured epoxy, acetone wipes, and contaminated absorbents should be collected as chemical waste unless EHS approves another route.
- TiO₂-contaminated gloves, wipes, or weighing boats should be collected in a labeled solid waste container according to institutional EHS instructions.
- Do not pour TiO₂, uncured PDMS, or epoxy down the drain.

9. Shipping and Storage Protocols

9.1. Storage

- Storage in room temperature: 20-25 °C
- Store in the enclosure box and avoid direct sunlight and prolonged UV exposure.
- Avoid freezing or refrigeration unless stability testing demonstrates no change in optical or mechanical properties.

9.2. Shipping protocol

- Place the phantom package in a sealed enclosure.
- Add cushioning such as foam inserts or bubble wrap to prevent movement.
- Place the primary container in a rigid secondary box.

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